College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 05

Focus: Inheritance, super, protected, polymorphism intro

Difficulty: Moderate

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Building an inheritance tree.

a. Write class `Person` with private `name`, a constructor, and `getName()`. **(6 marks)**

b. Write class `Teacher extends Person` with private `subject`, constructor calling `super`, and `display()`. **(10 marks)**

c. Explain what `super(name)` does in the subclass constructor. **(4 marks)**

Question 2 [20 Marks]

Protected access.

a. How does `protected` differ from `private` and `public`? **(6 marks)**

b. Give a short example where a subclass method reads a protected superclass field. **(8 marks)**

c. When would you still prefer private fields + getters over protected fields? **(6 marks)**

Question 3 [20 Marks]

Method overriding.

a. What is overriding? How is it different from overloading? **(6 marks)**

b. Write superclass method `speak()` and a subclass override that prints a different message. Include `@Override`. **(8 marks)**

c. What is printed by `Animal a = new Dog(); a.speak();` if `Dog` overrides `speak`? Why? **(6 marks)**

Question 4 [20 Marks]

Polymorphism - first ideas.

a. Define polymorphism in the context of OOP. **(4 marks)**

b. Explain why a list of `Vehicle` references can hold `Car` and `Bike` objects. **(6 marks)**

c. Write a small demo: array or list of superclass type, two subclasses, loop calling an overridden method. **(10 marks)**

Question 5 [20 Marks]

```
class Shape {
    public double area() { return 0; }
}
class Circle extends Shape {
    private double r;
    public Circle(double r) { this.r = r; }
    public double area() { return 3.14 * r * r; }
}
```

Study the code.

a. Identify the superclass, subclass, and overridden method. **(6 marks)**

b. What does `Shape s = new Circle(2); System.out.println(s.area());` print (approx)? **(4 marks)**

c. Rewrite `Shape.area` as `abstract` and make `Shape` an abstract class. Adjust `Circle` if needed. **(10 marks)**

Question 6 [20 Marks]

Library Phase 4/5 style thinking.

a. Suggest a superclass for different library items (book, DVD) and two subclasses with one extra field each. **(8 marks)**

b. Which behaviours belong in the superclass and which in subclasses? Justify briefly. **(6 marks)**

c. How does inheritance reduce duplicated code in borrow/return logic? **(6 marks)**

Question 7 [20 Marks]

Exceptions - awareness (additional topic).

a. What is an exception? What happens if it is not caught? **(4 marks)**

b. Write a `try/catch` around `Integer.parseInt` that prints a friendly message on failure. **(8 marks)**

c. Name three common exception types and one cause for each. **(6 marks)**

d. Why is checking `if (obj != null)` often better than catching `NullPointerException`? **(2 marks)**

******* END OF QUESTION PAPER *******